MODULE 1 CELL ANATOMY

Anatomy of the Cell.

CELL ANATOMY

The cell is the smallest living unit of the body. Every organelle in it is a structure with a job, and every one of those jobs has a disease that shows what happens when it fails.

COMPETENCIES W1-GENERALIZED-CELL, W1-PLASMA-MEMBRANE, W1-NUCLEUS, W1-ORGANELLE-ID

BY THE END

1. Identify the three major components of a cell and describe what each does.
2. Define semipermeable, permeable, and impermeable, with examples.
3. Name the structures of the nucleus and state the function of each.
4. Separate cytoplasm from cytosol.
5. Describe the plasma membrane and the fluid mosaic model.
6. List the components of the endomembrane system and trace a protein through it.
7. Describe each organelle and connect it to a clinical consequence when it fails.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The Cell

drsrennie-stack.github.io/new-build-bio4-solano/cell-concept-videos.html

General Structure of the Cell

Plasma membrane

A flexible, semipermeable boundary that encloses the cell, separates its contents from the external environment, and regulates the passage of substances. It is composed primarily of lipids and proteins with some carbohydrates, and it carries specialized structures such as receptors, ion channels, and transport proteins for communication and material exchange.

Nucleus

The largest organelle and the command center of the cell. It contains the genetic material, DNA, organized into chromosomes. It controls gene expression, cellular growth, and replication, and it is the site of transcription, the conversion of DNA to RNA.

Cytoplasm

The gel-like substance within the plasma membrane, excluding the nucleus, containing the organelles and the cytosol. It facilitates intracellular processes such as glycolysis and signal transduction, helps maintain cell shape, and enables the movement of materials.

Cell Permeability

THREE DEGREES OF PERMEABILITY

Term	Definition	Examples
Semipermeable	Allows selective substances to cross based on size, charge, and solubility. Facilitates passive diffusion for some molecules and relies on transport proteins for others	Oxygen, carbon dioxide, and small lipid-soluble molecules diffuse freely
Permeable	Permits the free passage of substances without restriction	Water through aquaporins, gases such as nitrogen
Impermeable	Blocks movement unless a specific mechanism carries the substance across	Proteins, polysaccharides, charged ions with no transporter

The Nucleus in Detail

Multinucleated cells have more than one nucleus. Skeletal muscle fibers and hepatocytes in the liver are the standard examples. **Anucleated cells** have none. Mature red blood cells are the example, and losing the nucleus is what makes room for hemoglobin.

Function. The nucleus stores the DNA that carries instructions for protein synthesis and cellular activities, plays a central role in cell division, growth, and metabolism, and regulates which genes are turned on or off in different cells. That last one is what allows differentiation.

STRUCTURES OF THE NUCLEUS

Structure	What it is and does
Nuclear envelope	A double-layered membrane separating the nucleus from the cytoplasm and maintaining a controlled environment
Nuclear pores	Protein-lined channels allowing exchange of materials such as RNA and proteins between nucleus and cytoplasm
Chromatin	A network of DNA and histone proteins that condenses into visible chromosomes during cell division
Chromosomes	Rod-shaped structures containing genes, responsible for passing genetic information during reproduction
Nucleoplasm	The viscous fluid within the nucleus supporting the chromatin and nucleolus
Nucleolus	A dense region where ribosomal RNA is synthesized and ribosomal subunits are assembled

Cytoplasm versus Cytosol

Cytoplasm is the entire contents of the cell excluding the nucleus, meaning the organelles suspended in cytosol. **Cytosol** is the aqueous component alone: water, ions, enzymes, and small organic molecules. The cytosol is where many metabolic reactions happen, glycolysis among them.

The Plasma Membrane in Detail

Function. A dynamic barrier that regulates interactions between the cell and its environment.

MEMBRANE COMPONENTS

Component	Arrangement and role
Phospholipid bilayer	Hydrophilic heads face outward toward the extracellular and intracellular fluids. Hydrophobic tails face inward, away from water
Proteins	Integral proteins span the bilayer, peripheral proteins bind to the surface. Roles: transport, enzymatic activity, signal transduction
Glycoproteins and glycolipids	Cell recognition and immune responses
Cholesterol	Provides stability and moderates membrane fluidity at varying temperatures

Fluid mosaic model. The membrane behaves as a fluid-like sea of phospholipids with proteins and carbohydrates floating within it. Proteins and lipids move laterally, which allows flexibility and dynamic interaction rather than a fixed wall.

The Endomembrane System

A coordinated network of organelles responsible for protein and lipid synthesis, modification, and transport.

- **Nuclear envelope**, protects the genetic material.
- **Rough ER**, synthesizes membrane-bound and secretory proteins.
- **Smooth ER**, synthesizes lipids, detoxifies chemicals, stores calcium.
- **Golgi apparatus**, modifies, sorts, and packages proteins and lipids.
- **Lysosomes**, digestive organelles containing hydrolytic enzymes.
- **Vesicles**, transport materials between organelles or to the plasma membrane.

Ribosomes

Function. The site of protein synthesis, where mRNA is translated into polypeptides. **Structure.** A large and a small subunit, each made of ribosomal RNA and protein.

- **Free ribosomes**, produce proteins for use inside the cell.
- **Bound ribosomes**, attached to the rough ER, producing proteins for secretion or for insertion into membrane.
- **Mitochondrial ribosomes**, produce proteins specific to mitochondrial function.

Endoplasmic Reticulum

ROUGH AND SMOOTH ER COMPARED

	Rough ER	Smooth ER
Structure	Studded with ribosomes	Lacks ribosomes
Function	Produces membrane proteins and secretory proteins	Synthesizes lipids, metabolizes carbohydrates, detoxifies drugs, stores calcium ions
Next stop	Sends vesicles to the Golgi apparatus	Supplies lipid and calcium to the rest of the cell

Golgi Apparatus

Function. Modifies, packages, and sorts proteins and lipids, tagging molecules for delivery to specific cellular locations. **Structure.** A series of flattened sacs called cisternae, with vesicles budding off.

Lysosomes

Function. Digestive organelles that break down macromolecules, damaged organelles, and pathogens. Two processes to know: **phagocytosis**, engulfing pathogens or debris, and **autophagy**, recycling of the cell's own components.

CLINICAL CORRELATION

In Tay-Sachs disease, a deficiency of a lysosomal enzyme causes toxic accumulation of lipids in nerve cells. The organelle that should be clearing material cannot, so the material builds up inside the cell until the cell fails.

Peroxisomes

Function. Break down fatty acids, amino acids, and hydrogen peroxide. They also detoxify alcohol in the liver and kidney, which is why those organs are peroxisome-rich.

Mitochondria

Function. The powerhouse of the cell, producing ATP through cellular respiration. **Structure.** A double membrane with inner folds called cristae, which multiply the surface area available for ATP production. Mitochondria contain their own DNA and ribosomes, so they produce some of their own proteins.

Cytoskeleton

Function. Provides structural support, facilitates intracellular transport, and enables cellular movement.

THREE CYTOSKELETAL COMPONENTS

Component	Made of	Role
Microfilaments	Actin	Cell shape and movement
Intermediate filaments	Various fibrous proteins	Mechanical support, resisting tension
Microtubules	Tubulin	Tubular structures forming centrioles, cilia, and flagella

Centrioles and Centrosomes

Centrioles are short cylindrical structures organized in a 9 plus 0 microtubule arrangement. The **centrosome** is the microtubule-organizing center. Both are essential for forming the mitotic spindle during cell division.

Cilia and Flagella

Cilia are short projections that move substances across the cell surface, such as mucus in the respiratory tract. **Flagella** are long projections that propel the cell itself, as in sperm. Both use a 9 plus 2 microtubule arrangement, and that arrangement is what makes movement possible.

HOLD ONTO THIS

9 plus 0 does not move, it organizes. 9 plus 2 moves. Centrioles are 9 plus 0. Cilia and flagella are 9 plus 2.

THE ORGANELLE POINTS TO THE DISEASE

Read this list backward when you are given a case. A cell that cannot break down lipids has a lysosome problem. A cell that cannot make enough ATP has a mitochondrial problem, and the tissues that fail first are the ones with the highest energy demand: muscle, brain, kidney. A cell that cannot clear mucus has a ciliary problem, and the presentation is recurrent respiratory infection. Structure first, then the consequence.